

Lecture 04

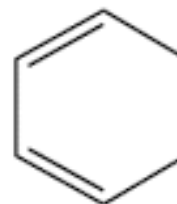
Mass Spectroscopy

PART - II



MS- Fragmentation Process

Eg: 1,3 - Cyclohexadiene



- Fragmentation is initiated by electron impact
- Major driving force is the cation-radical character
- A- delocalized structure with one less electron than the 1,3 - Cyclohexadiene
- Fragmentation of odd- electron molecular ion may occur by homolytic or heterolytic cleavage of a single bond
 - Homolytic cleavage
 - Heterolytic cleavage

- Structures A, B, and C in Figure 1.13, for example, represent the molecular ion of cyclohexadiene.
- Compound A is a delocalized structure with one less electron than the original uncharged diene; both the electron and the positive charge are delocalized over the π system. Since the electron removed to form the molecular ion is a π electron, other structures, such as B or C (valence bond structures) can be used. Structures such as B and C localize the electron and the positive charge and thus are useful for describing fragmentation processes.

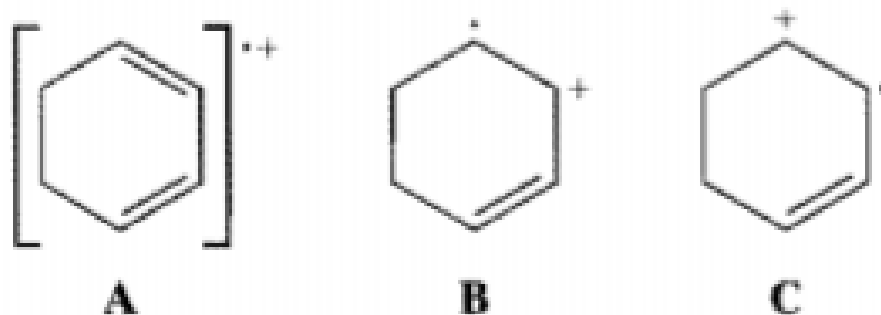
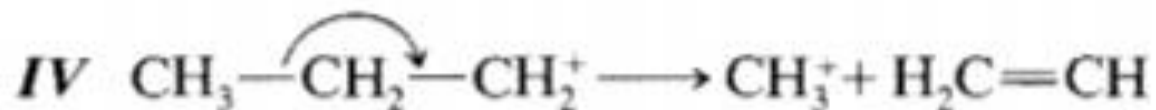
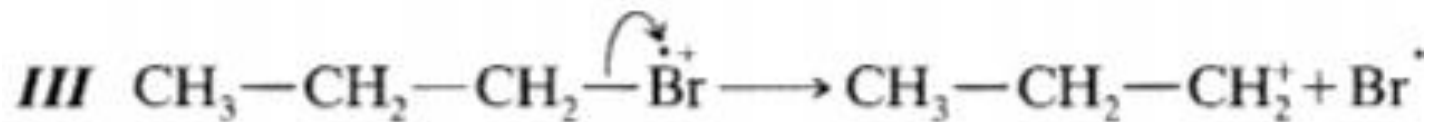
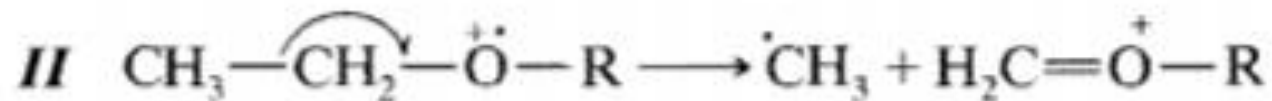
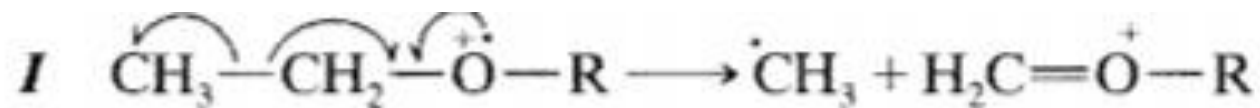


FIGURE 1.13 Different representations of the radical cation of cyclohexadiene.

- Fragmentation is initiated by electron impact. Only a small part of the driving force for fragmentation is energy transferred as the result of the impact.
- The major driving force is the cation-radical character that is imposed upon the structure.
- Fragmentation of the odd-electron molecular ion (radical-cation, $M^{\cdot+}$) may occur by homolytic or heterolytic cleavage of a single bond. In homolytic cleavage (Scheme 1.3, I) each electron "moves" independently as shown by a (single-barbed) fishhook: the fragments are an even-electron cation and a free radical (odd electron).
- To prevent clutter, only one of each pair of fishhooks need be shown (Scheme 1.3 II). In heterolytic cleavage a pair of electrons "move"



- In the absence of rings, the most predominant fragments in a mass spectrum are even-electron cation formed by a single cleavage.
- Further fragmentation of an even-electron cation usually results in another even-electron and an even electron neutral molecule or fragment

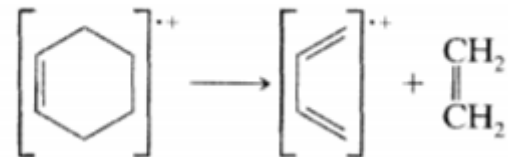
Rules

- The relative height of the molecule ion peak is greatest for the straight-chain compound and decreases as the degree of branching increases
- The relative height of the molecular ion peak usually decreases with increasing molecular weight in a homologous series. Fatty esters appear to be an exception.
- Cleavage is favored at alkyl-substituted carbon atoms: the more substituted, the more likely is cleavage. This is a consequence of the increased stability of a tertiary carbocation over a secondary, which in turn is more stable than a primary.

Cation stability order: $\text{CH}_3^+ < \text{R}_2\text{CH}_2^+ < \text{R}_3\text{CH}^+ < \text{R}_3\text{C}^+$

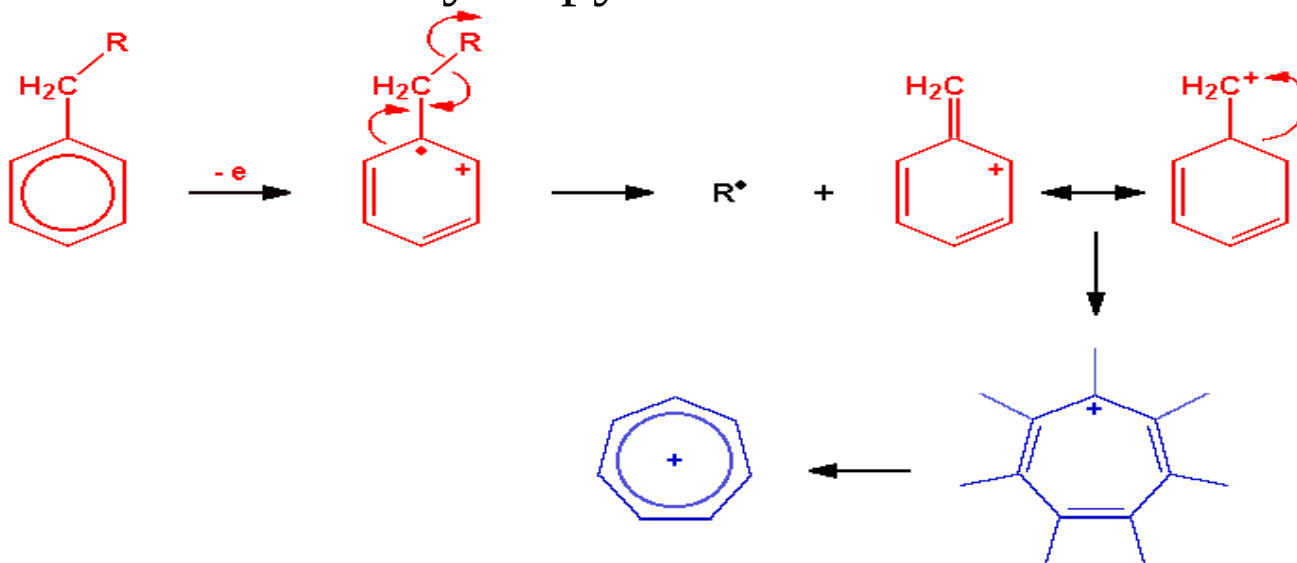
- Saturated rings tend to lose alkyl side chain at the α bond. The (+) charge tends to stay with the ring fragment

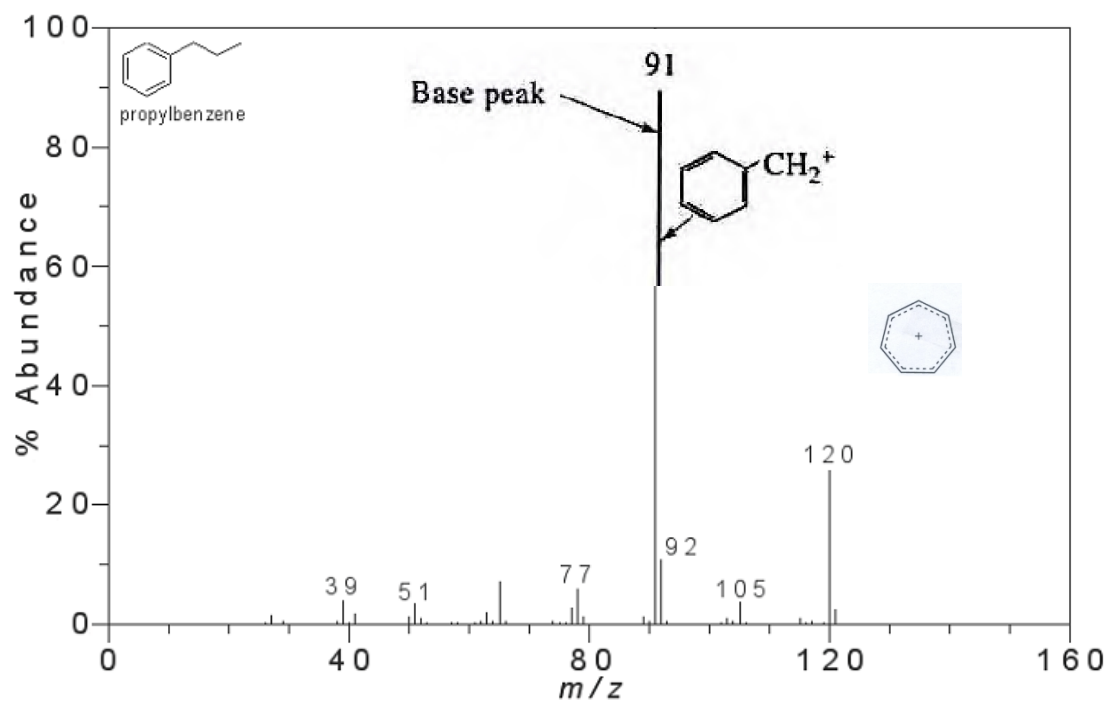
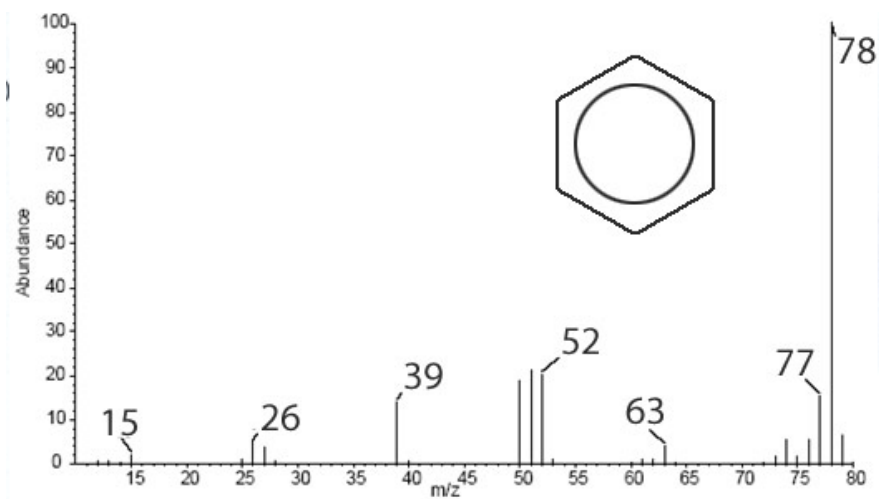
(eg: Retro - Diels -Alder reaction)



Arenes tend to cleave at benzylic position

- In alkyl substitutes, cleavage of aromatic compounds are very probable at the β bond to the ring giving the resonance stabilized benzene ion or more likely tropylium ion



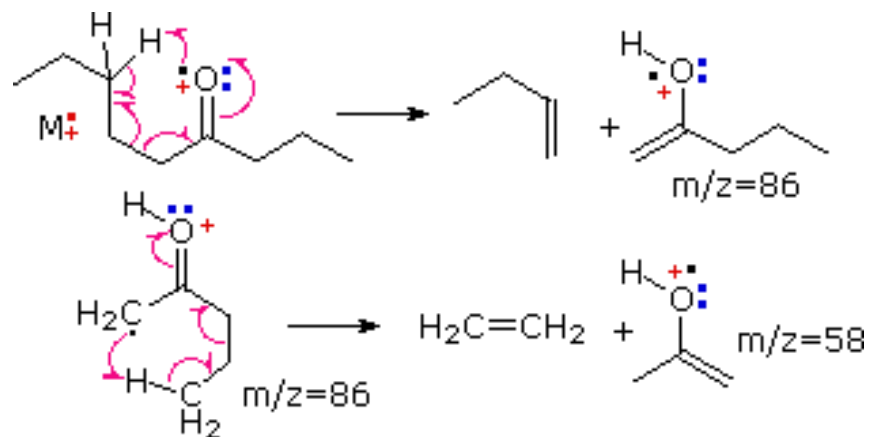
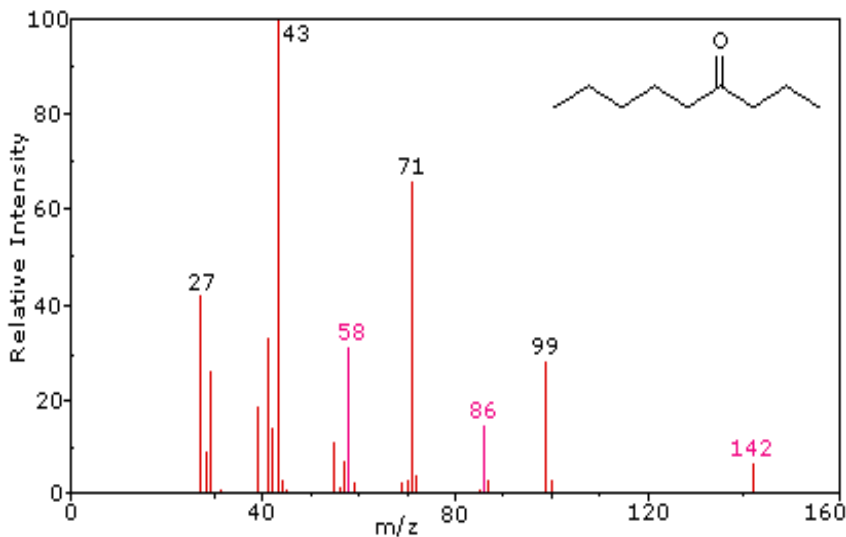


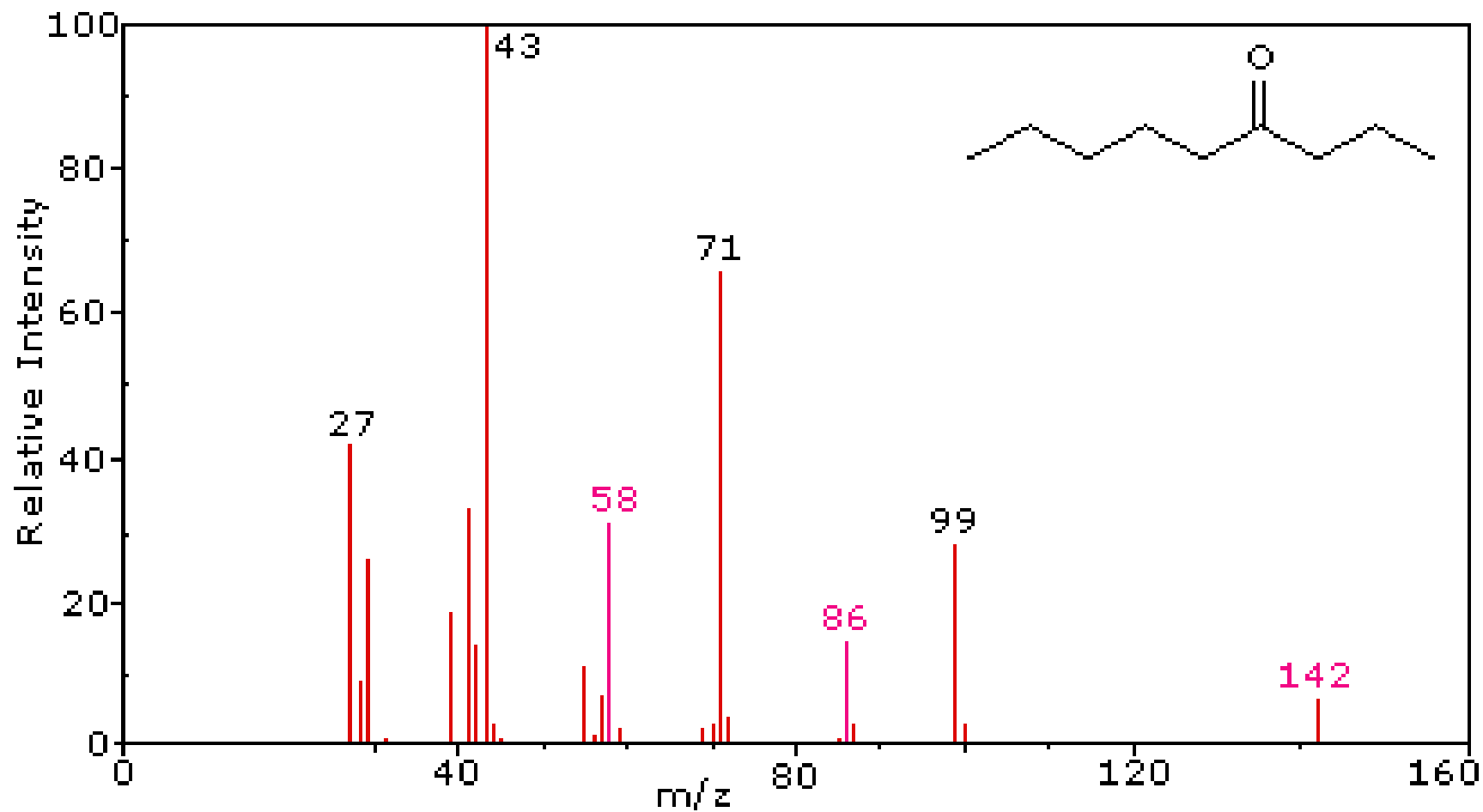
- The C-C bond next to heteroatom are frequently cleaved leaving the charge on the fragment containing the heteroatom whose non-bonding electrons provide resonance stabilization.

McLafferty rearrangement

- The odd-electron fragment ions at $m/z = 86$ and 58 are the result of a McLafferty rearrangement, involving the larger alkyl chain, and a subsequent loss of ethene (the "double-McLafferty" rearrangement).
- Alpha-cleavage leads to the $m/z = 99$, 71 and 43 ions. The charge is apparently distributed over both fragments :

Eg = 4 - Nonanone



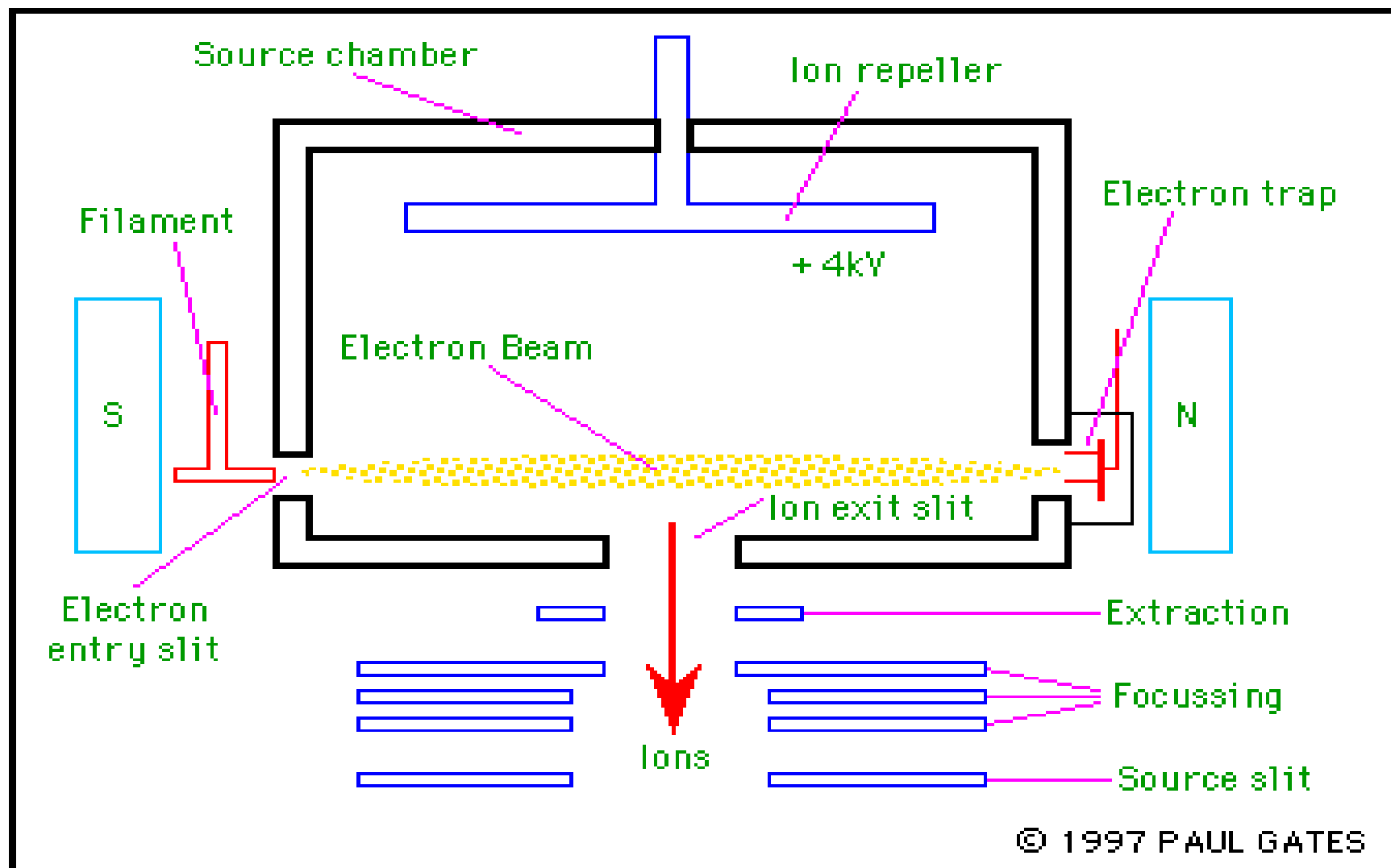


Ionization modes

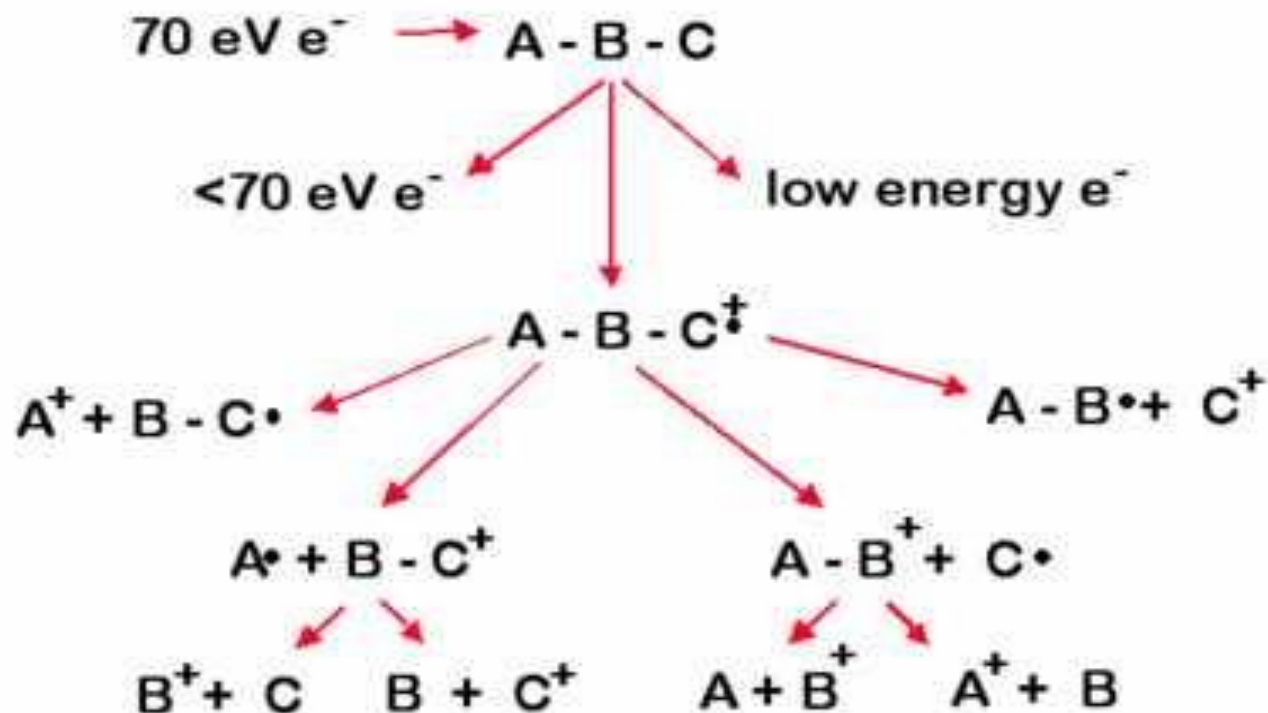
1. Electron-impact Ionization (EI)
2. Chemical Ionization or Positive Chemical Ionization (CI)
3. Negative Chemical Ionization (NCI)

1. Electron-impact Ionization (EI)

- High energy electrons (70 eV) from the filament (thermionic emission from a tungsten or rhenium filament) will hit the neutral molecule from the sample
- This impact will cause an electron to be expelled from the neutral molecule, generating a molecular ion (parent ion)
- The molecular ion has high internal energy cause it to fragment into smaller ions (daughter ions)



Electron impact process



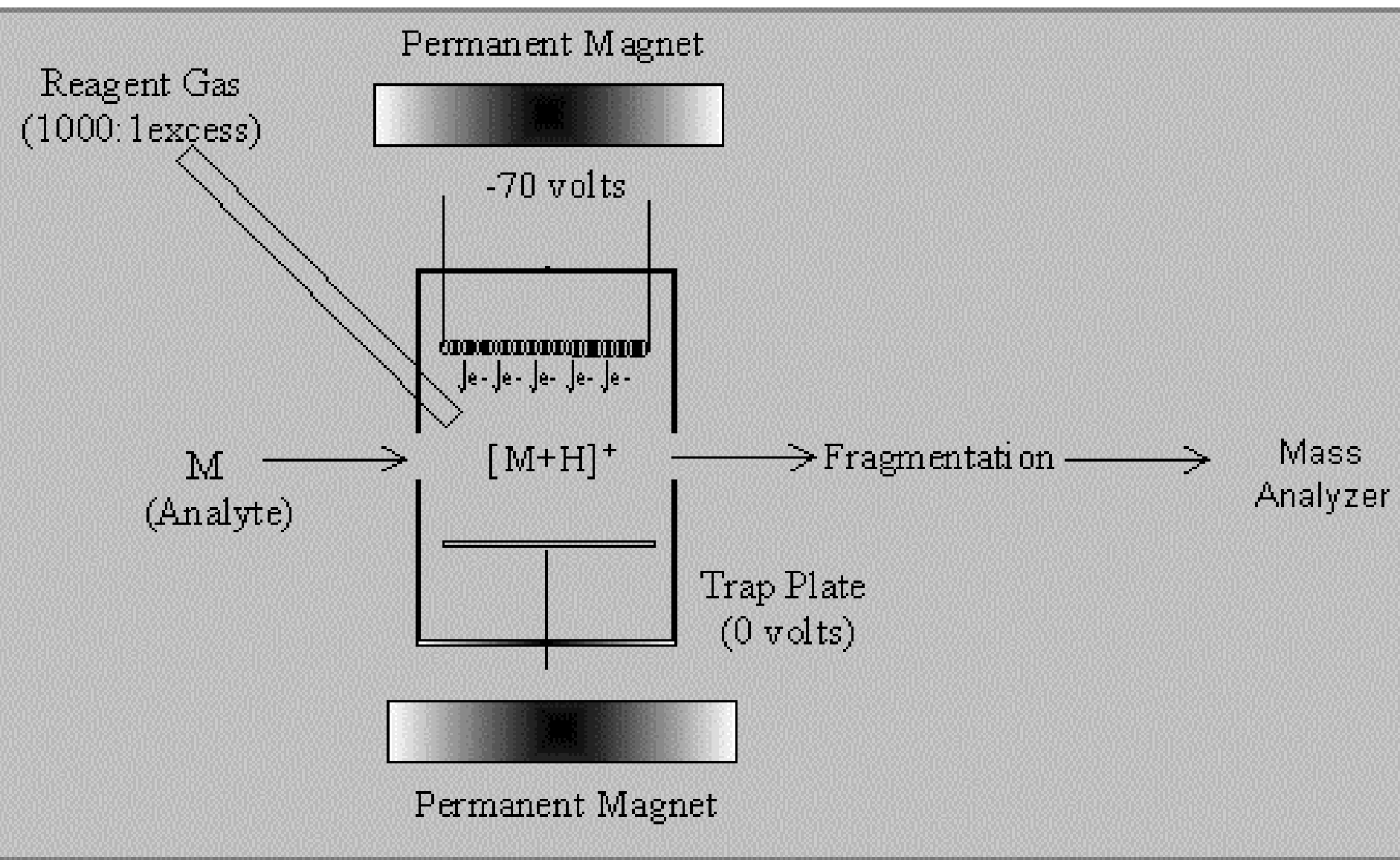
2. Chemical Ionization (CI)

- High energy electrons will hit the reagent gas (usually methane, ethane or ammonia) to form reagent ions.
- These reagent ions then undergo protonation reaction with the analytes
- The positive ions are analyzed and detected

3. Negative Chemical Ionization (NCI)

- High energy electrons will hit the reagent gas (usually methane, ethane or ammonia) to form reagent ions.
- The low energy electrons then interact with the analytes (usually with a more electronegative elements) to form negative ions
- The negative ions are analyzed and detected

Chemical Ionization

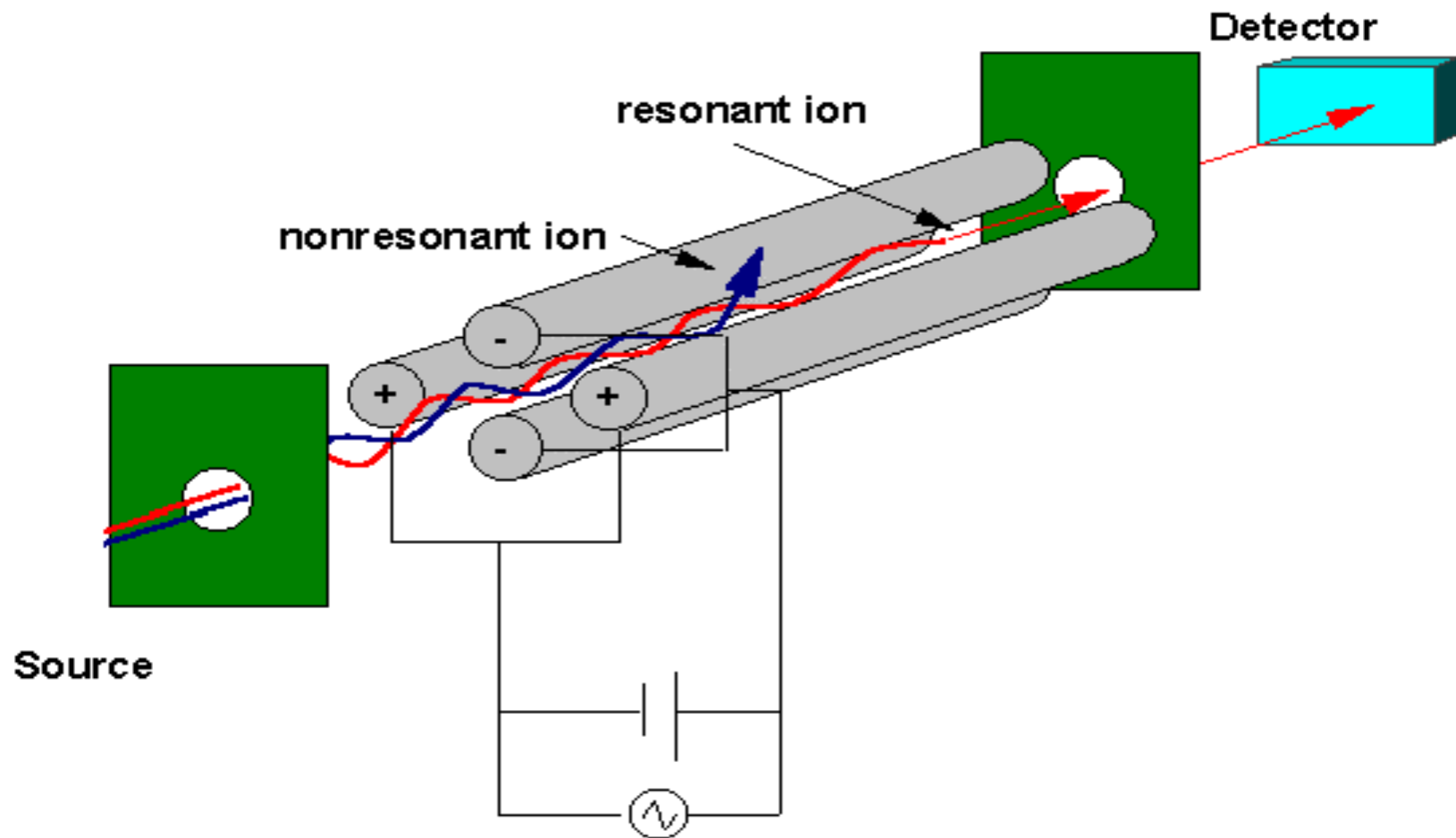


Detection of Ions

1. Quadrupole Mass Analyzer

- Consists of four parallel rods or poles through which the ions being separated are passed.
- Depending on the produced electric field, only ions of a particular m/z will be focused on the detector, all the other ions will be deflected into the rods.
- By varying the strengths and frequencies of electric fields, different ions will be detected thus making the mass spectrum.

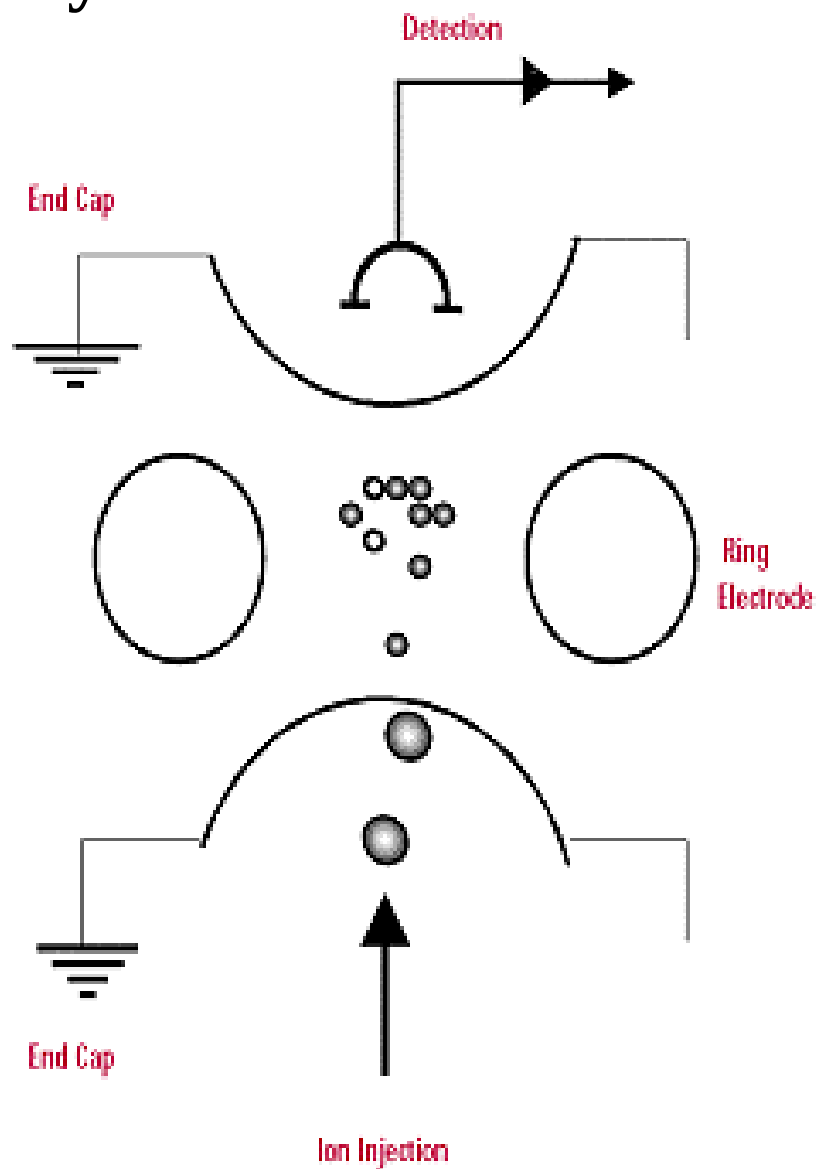
Quadrupole Mass Analyzer



Ion trap mass Analyzer

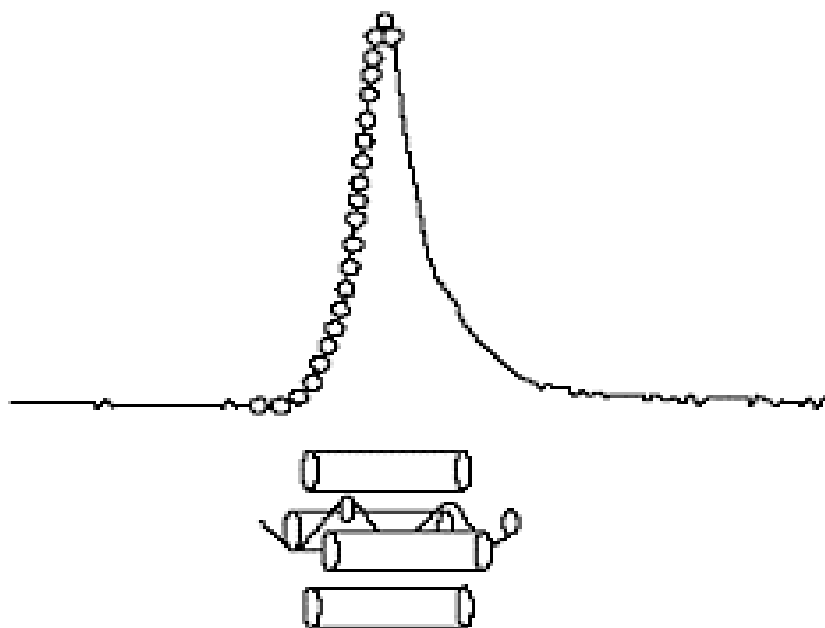
- Store the ions in a device consisting of a ring electrode and two end cap electrodes.
- The ions are stabilized in the trap by applying a RF voltage on the ring electrode.
- For maximum efficiency, the ions must be focused near the centre where the trapping fields are closest to the ideal and the least distorted - maximizing resolution and sensitivity.
- This is achieved by introducing a damping gas (99.998% helium) that collisionally cools injected ions, damping down their oscillations until they stabilize.

Ion trap mass Analyzer



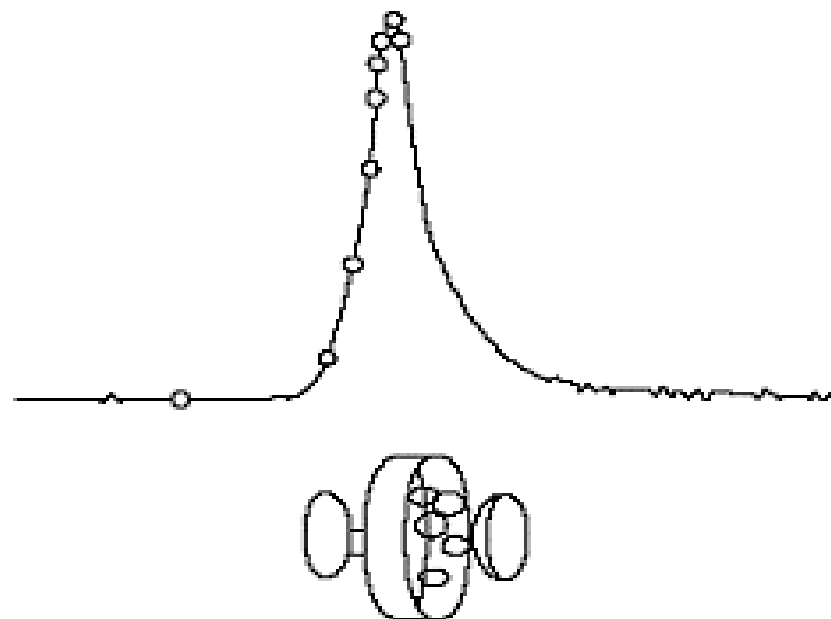
Triple Quadrupole

Data points are evenly spaced across the entire peak



Ion Trap

Data points are obtained most frequently at the top of the peak



Time-of-flight mass spectrometry (TOF-MS)

- The sample ions are generated in a source zone, s , of the instrument, by whatever ionization method is being employed.
- A potential, (V_s - the source extraction) is applied across the source to extract and accelerate the ions from the source into the field-free 'drift' zone of the instrument, d .
- The basic components of TOF-MS. The larger the ion, the slower its velocity and thus the longer it takes to traverse the field-free drift zone. The field in the field-free zone, E_d , is zero at all times

Basic formula for TOF mass analysis is given by the equation:

Where:

$$\frac{m_i}{z_i} = 2eE l_s \left(\frac{t_i}{l_d} \right)^2$$

m_i = mass of analyte ion z_i = charge on analyte ion

E = extraction field t_i = time-of-flight of ion

l_s = length of the source

l_d = length of the field-free drift region

e = electronic charge (1.6022×10^{-19} C)

Time-of-flight mass spectrometry (TOF-MS)

